

## 732A96/TDDE15 ADVANCED MACHINE LEARNING

EXAM 2026-01-07

### TEACHER

Jose M. Peña. He will answer questions on the phone.

### GRADES

- For 732A96 (A-E means pass):
  - A=19-20 points
  - B=17-18 points
  - C=14-16 points
  - D=12-13 points
  - E=10-11 points
  - F=0-9 points
- For TDDE15 (3-5 means pass):
  - 5=18-20 points
  - 4=14-17 points
  - 3=10-13 points
  - U=0-9 points

In each question, full points requires clear and well motivated answers and commented code.

### INSTRUCTIONS

- This is an individual exam. No help from others is allowed. No communication with others is allowed. Answers to the exam questions may be sent to Urkund.
- This is an anonymous exam. Do not write your name on it.
- The answers to the exam should be submitted in a single PDF file. You can make a PDF from LibreOffice (similar to Microsoft Word). You can also use Markdown from RStudio (no support is provided though). Include important code needed to grade the exam (inline or at the end of the PDF file).

### ALLOWED HELP

Everything on the course web page. Your individual and group solutions to the labs. This help is available on the corresponding directories of the exam system.

## 1. PROBABILISTIC GRAPHICAL MODELS (7 P)

In this exercise, no implementation is required. In the course, we have seen how to read the independencies represented by a DAG with the help of the d-separation criterion. An alternative is the moralization criterion. To check if a DAG  $G$  represents that  $X$  is independent of  $Y$  given  $Z$ , first, identify the ancestral set  $S$  of all nodes in  $X \cup Y \cup Z$ , i.e., all the nodes in  $X \cup Y \cup Z$  plus all the nodes that have a directed path to a node in  $X$ ,  $Y$  or  $Z$ . Second, restrict the original DAG  $G$  to the subgraph containing only the nodes in  $S$  and their corresponding edges, which we denote as  $G_S$ . Third, construct the moral graph  $G_S^m$  of  $G_S$  by adding an edge between any two non-adjacent nodes in  $G_S$  that are both parents of the same child node (sometimes described as "marrying unmarried parents"), and then dropping all edge directionalities. Finally, if all paths between  $X$  and  $Y$  in  $G_S^m$  are blocked by  $Z$ , then the original DAG  $G$  represents that  $X$  is independent of  $Y$  given  $Z$ .

Apply the moralization criterion to determine if  $A$  is independent of  $B$  given  $\{D, E\}$  in the DAG  $G = \{E \rightarrow A, A \rightarrow C, C \rightarrow D, C \leftarrow B, B \leftarrow E\}$ . Sketch a formal proof that moralization and d-separation are equivalent. Answer the following questions: Under what conditions does a DAG have a Markov equivalent undirected graph, i.e., an undirected graph that encodes exactly the same set of conditional independencies as the DAG? How many undirected graphs are Markov equivalent to a given undirected graph?

## 2. REINFORCEMENT LEARNING (6 P)

Q-learning tends to overestimate action values because it always uses the maximum over the estimated q-values in the updating rule:

$$q(s, a) \leftarrow q(s, a) + \alpha(r + \gamma \max_{a'} q(s', a') - q(s, a)).$$

Double Q-learning is a modification of Q-learning that addresses this overestimation by maintaining two separate q-value tables,  $q_1$  and  $q_2$ . At each step:

- With probability 0.5, it updates  $q_1$  using  $q_2$  to evaluate the greedy next action:

$$q_1(s, a) \leftarrow q_1(s, a) + \alpha(r + \gamma q_2(s', \arg \max_{a'} q_1(s', a')) - q_1(s, a)).$$

- Otherwise, it updates  $q_2$  using  $q_1$  analogously:

$$q_2(s, a) \leftarrow q_2(s, a) + \alpha(r + \gamma q_1(s', \arg \max_{a'} q_2(s', a')) - q_2(s, a)).$$

The behavior policy for action selection remains  $\epsilon$ -greedy based on  $q_1(s, a) + q_2(s, a)$ .

You are asked to modify your lab solution to implement Double Q-learning. Run it for 10000 episodes in the environment A with the same parameter values as in the lab. Plot the moving average of the rewards obtained in those 10000 episodes. Compare the results with those from Q-learning. Comment the results. Answer the following question: Why does Double Q-learning reduce the overestimation bias compared with Q-learning?

## 3. GAUSSIAN PROCESSES (7 P)

In the course, we have seen the equations for the posterior mean and covariance of a GP for regression. These equations can be implemented directly, or indirectly as in Algorithm 2.1. When implemented directly, it is sometimes more convenient to implement an approximation to them. For instance, Nyström approximation consists in selecting some points  $Z$  of the dataset at random (so-called inducing points), and then approximate  $K(X, X)$  as follows:

$$K(X, X) \approx K(X, Z)K(Z, Z)^{-1}K(Z, X).$$

This intuitively means that the covariance between two points is approximated as the covariance between the projections of the points onto the space spanned by the inducing points.

In the original equations for the posterior mean and covariance of a GP for regression,  $K(X, X)$  appears in the term

$$[K(X, X) + \sigma_n^2 I]^{-1}$$

which, under Nyström approximation, should be replaced by

$$[K(X, Z)K(Z, Z)^{-1}K(Z, X) + \sigma_n^2 I]^{-1}.$$

This matrix inversion can efficiently be computed with the help of Woodbury matrix identity (a.k.a. matrix inversion lemma):

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U[C^{-1} + VA^{-1}U]^{-1}VA^{-1}$$

if we simply let  $A = \sigma_n^2 I$ ,  $U = K(X, Z)$ ,  $C = K(Z, Z)^{-1}$  and  $V = K(Z, X)$ . Note that  $A^{-1} = I/\sigma_n^2$ .

A common improvement over pure low-rank approximations (such as Nyström) is to add a diagonal correction that preserves the exact marginal variances at the training points. This leads to a low-rank plus diagonal approximation of  $K(X, X)$  as follows:

$$K(X, X) \approx Q + \Lambda$$

where

$$Q \approx K(X, Z)K(Z, Z)^{-1}K(Z, X)$$

and

$$\Lambda = \text{diagonal}(K(X, X) - Q).$$

The operation  $\text{diagonal}(M)$  returns the diagonal matrix whose diagonal entries are the diagonal entries of  $M$ .

In the original equations for the posterior mean and covariance of a GP for regression,  $K(X, X)$  appears in the term

$$[K(X, X) + \sigma_n^2 I]^{-1}$$

which, under the low-rank plus diagonal approximation, should be replaced by

$$[Q + \Lambda + \sigma_n^2 I]^{-1}.$$

This matrix inversion can efficiently be computed with the help of Woodbury matrix identity much as before by now letting  $A = \Lambda + \sigma_n^2 I$ . Note that  $A$  is diagonal, and thus  $A^{-1}$  is diagonal too. Specifically,  $(A^{-1})_{ii} = 1/A_{ii}$ .

Your task is to derive and implement the equations for the posterior mean and covariance of a GP under (i) Nyström approximation, and (ii) the low-rank plus diagonal approximation. Use 100 randomly sampled data points as inducing points. When calling the function `solve`, use the parameter `tol = 0` to avoid numerical problems. Run your code on the Tullinge dataset to predict temperature from time, and compare the results of the two approximations with those you obtained in the lab. Use the template below. Use the local copy of the dataset in the `help` folder. Use the same setting as in the lab (every 5th observation, squared exponential kernel,  $\sigma_F = 20$  and  $\ell = 100$ ).

```
SEKernel <- function(x1,x2){
  n1 <- length(x1)
  n2 <- length(x2)
  K <- matrix(NA,n1,n2)
  for (i in 1:n2){
```

```
      K[,i] <- (sigmaF^2)*exp(-0.5*((x1-x2[i])/1)^2 )
    }
    return(K)
}
```

```
# Your code here.
```

```
Kxs <- SEKernel(x,xs)
Kss <- SEKernel(xs,xs)
Kxz <- SEKernel(x,z)
Kzz <- SEKernel(z,z)
```

```
# Your code here.
```